

Exercise Sheet 2

Dependence & Bivariate data exploration

Use of AI You may use AI tools to help draft code, but your final answers should show your own judgment and explain why the results look like that.

Setup in Python Use `numpy`, `pandas`, `matplotlib`, `scipy`, `seaborn`, `yfinance`, and `pyvinecopulib`. Use the two tickers SPY (S&P 500 ETF) and BAC (Bank of America Corporation). Use the time window

`start_date = 2000-01-01,` `end_date = 2025-12-31.`

If the API returns no record on `end_date`, use the closest previous trading day.

Notation For asset $j \in \{1, 2\}$, let $P_d^{(j)}$ denote the chosen daily price on trading day d :

$$P_d^{(j)} := \begin{cases} \text{Adj Close for asset } j \text{ on day } d, & \text{if available,} \\ \text{Close for asset } j \text{ on day } d, & \text{otherwise.} \end{cases}$$

Do not mix `Close` and `Adj Close` within an asset. Use trading days only; do not fill non-trading days; drop missing prices before computing returns. For this exercise, convert the daily price series into a monthly price series by taking the last available trading day in each calendar month. Let $S_m^{(j)}$ denote the monthly price for month m and asset j .¹ Define the one period monthly log-return by

$$r_m^{(j)} := \log S_m^{(j)} - \log S_{m^-}^{(j)},$$

where m^- denotes the previous month in the monthly series for asset j .

Let $\mathcal{T}_j$ denote the set of months on which $r_m^{(j)}$ is available, and let $\mathcal{T} := \mathcal{T}_1 \cap \mathcal{T}_2 = \{m_1, \dots, m_n\}$ be the common set of return months, listed in increasing order. Define the bivariate sample by

$$\mathbf{R}_t := \begin{pmatrix} X_t \\ Y_t \end{pmatrix} := \begin{pmatrix} r_{m_t}^{(1)} \\ r_{m_t}^{(2)} \end{pmatrix}, \quad t = 1, \dots, n.$$

The associated data matrix is

$$\mathbf{R} := \begin{pmatrix} X_1 & Y_1 \\ \vdots & \vdots \\ X_n & Y_n \end{pmatrix} \in \mathbb{R}^{n \times 2}.$$

¹To work with copula models for financial time series, one would first model and remove serial dependence and conditional heteroskedasticity, and then fit the copula to filtered innovations or standardized residuals. In this sheet, we use monthly returns as an approximation that reduces, but does not eliminate, serial dependence.

Problem 1 Download price data for SPY and BAC over the given time window:

- State the chosen price column for each asset, construct the bivariate return sample $\{\mathbf{R}_t\}_{t=1}^n$, and report n together with the first and last months.
- Generate a scatterplot of the sample $\{(X_t, Y_t) : t = 1, \dots, n\}$,
- Explain what the scatterplot tells you that two separate univariate plots of $\{X_t\}_{t=1}^n$ and $\{Y_t\}_{t=1}^n$ cannot tell you.
- Compute the empirical joint lower tail event frequency $\frac{1}{n} \sum_{t=1}^n \mathbf{1}\{X_t \leq q_{0.05}^X, Y_t \leq q_{0.05}^Y\}$, where $q_{0.05}^X$ and $q_{0.05}^Y$ are the empirical 5% quantiles of the samples $\{X_t\}_{t=1}^n$ and $\{Y_t\}_{t=1}^n$, respectively. Compare this value with $0.05^2 = 0.0025$, the value one would expect under independence of the two lower tail events.

Problem 2 For the sample $\{\mathbf{R}_t\}_{t=1}^n$:

- Compute the sample Pearson correlation and the sample Kendall's τ of $\{(X_t, Y_t) : t = 1, \dots, n\}$ and interpret the estimated dependence strength among the assets.
- Apply the strictly increasing transformation $g(x) = \text{sign}(x) \log(1 + 250|x|)$. Compare the two dependence measures for the original sample $\{(X_t, Y_t)\}_{t=1}^n$ and the transformed sample $\{(g(X_t), g(Y_t))\}_{t=1}^n$.

Problem 3

- Fit a bivariate Gaussian model $\mathcal{N}_2(\hat{\boldsymbol{\mu}}, \hat{\boldsymbol{\Sigma}})$ by maximum likelihood estimation. Estimate the parameters

$$\hat{\boldsymbol{\mu}} = \begin{pmatrix} \hat{\mu}_X \\ \hat{\mu}_Y \end{pmatrix}, \quad \hat{\boldsymbol{\Sigma}}_{\text{MLE}} = \begin{pmatrix} \hat{\sigma}_X^2 & \hat{\rho} \hat{\sigma}_X \hat{\sigma}_Y \\ \hat{\rho} \hat{\sigma}_X \hat{\sigma}_Y & \hat{\sigma}_Y^2 \end{pmatrix}.$$

- Generate an i.i.d. sample $\{\tilde{\mathbf{R}}_t\}_{t=1}^n$ of size n from the fitted bivariate Gaussian model. Compare the empirical sample $\{\mathbf{R}_t\}_{t=1}^n$ to the simulated sample $\{\tilde{\mathbf{R}}_t\}_{t=1}^n$ on the same scale with a scatter plot.
- Analyze/diagnose the Gaussian fit using marginal Q-Q plots for $\{X_t\}_{t=1}^n$ and $\{Y_t\}_{t=1}^n$ against their fitted Gaussian margins.
- Based on the diagnostics in (b)–(c), argue whether the fitted bivariate Gaussian is adequate.

Problem 4

- a) Construct pseudo-observations $U_t := \frac{\text{rank}(X_t)}{n+1}$ and $V_t := \frac{\text{rank}(Y_t)}{n+1}$ on the copula scale (in the unit square). Plot the sample $\{(U_t, V_t) : t = 1, \dots, n\}$.
- b) Explain what is lost and what is retained when $\{\mathbf{R}_t\}_{t=1}^n$ is replaced by $\{(U_t, V_t)\}_{t=1}^n$.
- c) Using `pyvinecopulib` library, fit
 - i) a Gaussian copula,
 - ii) a t -copula,
 - iii) a Gumbel copula, allowing its rotations,
 - iv) an independence copula.

Estimate the parameters and analyze the corresponding maximized pseudo log-likelihood and BIC. For the Gumbel family, also report the selected rotation in $\{0^\circ, 90^\circ, 180^\circ, 270^\circ\}$. Select the family–rotation combination with the smallest BIC.

- d) Use empirical marginal quantile functions to transform a sample to the return scale (using the copula fitted in c) with the best BIC value). Generate a scatterplot of
 - i) the observed sample $\{\mathbf{R}_t\}_{t=1}^n$,
 - ii) the fitted bivariate Gaussian sample from Problem 3,
 - iii) the selected copula sample on the return scale.

Discuss your findings.

- e) For each of the three samples in (d), construct the equally weighted portfolio loss

$$L_t = -\frac{X_t + Y_t}{2}.$$

Compute its empirical 95% VaR and ES. Which fitted model gives values closest to those obtained from the observed sample? Briefly explain why.

Problem 5 Let U and V be random variables with $\text{Unif}(0, 1)$ marginal distributions and let $C(u, v) := P(U \leq u, V \leq v)$ denote their copula.

- a) Prove that, for every $u, v \in [0, 1]$, $\max\{u + v - 1, 0\} \leq C(u, v) \leq \min\{u, v\}$.
- b) Show that both bounds are attainable by considering separately $V = U$ and $V = 1 - U$. Derive the copula in each case.